Graphic Era HILL UNIVERSITY

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End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: B. TECH, ME, III SEM
Name of the Paper: HEAT TRANSFER
Paper Code: TME 501
Time: 3 hour

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks) CO 1

a. Describe critical radius of insulation? Explain its importance in engineering problems. Write the relation for critical radius of cylinder and sphere.

b. Illustrate Biot and Fourier number.

c. Given: Two large aluminium (k = 240 W/mK), each 2 cm thick, with 10 mm surface roughness are placed in contact at 105 N/m2 pressure (Fig. Ex. 1.12) with the outside surface temperatures of 390 °C and 406°C. The thermal contact resistance is 2.75×10^{-4} m^2K/W. To find: (i) The heat flux, (ii) the temperature drop due to contact resistance, and (iii) the contact temperatures.

Q2. (2X10=20 Marks) CO 2

a. Derive an expression for heat transfer in extended surfaces of uniform cross section without heat generation.

b. A steel rod (k = 32W / °C) 12 mm in diameter and 60 mm long, with an insulated end, is to be used as a spine. It is exposed to surrounding with a temperature of 60 ° C and a heat transfer coefficient of 55W / m^2 °C the temperature at the base of fin is 95 ° C Determine:

(i) The fin efficiency.
(ii) Temperature at the edge of the spine.
(iii) The heat dissipation.

c. A plane wall with surface temperature of 350 °C is attached with straight rectangular fins (k 235 W/m*K). The fins are exposed to an ambient air condition of 25°C and the convection heat transfer coefficient is 154W/(m)*K Each fin has a length of 50 mm, a base of 5 mm thick and a width of 100 mm. Determine the efficiency, heat transfer rate, and effectiveness of each fin.

Q3. (2X10=20 Marks) CO 3

a. Analyze laminar and turbulent velocity boundary layers with the help of suitable flow diagram and Reynold's number.

b. Explain empirical correlation of free convection in vertical flat plate.

c. Given: Nitrogen gas at 0°C is flowing over a 1.2 m long, 2 m wide plate maintained at 80°C with a velocity of 2.5 m/s. For nitrogen, $\rho = 1.142$ kg/m^3, $c_p = 1.04$ kJ/kgK, $n = 15.63 \times 10^{-6}$ m^2/s and k = 0.0262 W/mK . To find: (a) The average heat transfer coefficient and (b) the total heat transfer from the plate.

Q4. (2X10=20 Marks) CO 4

a. Define Shape Factor. Derive an expression for the shape factor in case of radiation spherical ball inside a hollow rectangular box.

b. Calculate the net radiant heat exchanger per m^2 area for two large parallel plates at temperatures of 427 °C and 27 °C respectively. $\varepsilon_{hot\ plate} = 0.9$ and $\varepsilon_{cold\ plate} = 0.6$. If a polished aluminium shield is placed between them, find the percentage reduction, in the heat transfer, $\varepsilon_{shield} = 0.4$.

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c. A solid sphere of 1 m diameter at 600 K is kept in an evacuated, long, equilateral triangular enclosure whose sides are 2 m long. The emissivity of the sphere is 0.45 and the temperature of the enclosure is 420 K. If heat is generated uniformly within the sphere at a rate of 3100 W, determine (a)the view factor from the enclosure to the sphere and (b) emissivity of the enclosure.

Q5. (2X10=20 Marks) CO 5

a. Illustrate and derive overall heat transfer coefficients.

b. Describe fouling? what are the various factors by which fouling is affected.

c. Steam at atmospheric pressure enters the shell of a surface condenser in which the water flows through a bundle of tubes of diameter 25 mm at the rate of 0.05 kg/s. The inlet and outlet temperatures of water are 15 °C and 70 °C respectively. The condensation of steam takes place on the outside surface of the tube. If the overall heat transfer coefficient is 230W /m^2°C, calculate the following, using NTU method :

(i) The effectiveness of the heat exchanger.

(ii) The length of the tube.

(iii) The rate of steam condensation.

Take the latent heat of vaporisation at 100 ° C = 2257kJ / k * g